Mode S & ADS-B Statistics

What the receiver hears from the aircraft, and how to read and use the Statistics page

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In short. Every wind, temperature and windshear value of MODE-S Wind comes from radio messages that the aircraft transmit on 1 090 MHz: replies to the interrogations of the ground radars and the wide-area multilateration (WAM) network, and the aircraft's own ADS-B broadcasts. The Statistics page counts these messages as the receiver decodes them and shows, refreshed every 10 seconds:

- **what arrives** — messages per second, the downlink formats, the Comm-B registers and the ADS-B message types;
- **how often each aircraft answers** — the reply cadence of the registers that give the wind, per flight phase: the sample rate behind every wind and windshear value;
- **who is asking** — the ground interrogators identified from the all-call replies;
- **what it gives** — the winds computed per minute, and the positions in the Radarcape aircraft list;
- **how well the receiver hears** — the signal strength of every message and the range rings, the coverage per direction and altitude;
- **how the receiver itself is** — the Radarcape's own status: GPS lock and timestamps, GPS module temperature, uptime and settings;
- **whether the system is healthy** — a green, amber or red dot for each important part of MODE-S Wind, from the receiver feed to the Raspberry Pi itself.

Section 1 explains Mode S and ADS-B as far as needed to read the page, Section 2 the page itself, and Section 3 what can be analysed with it.

A passive view. The receiver hears the aircraft (1 090 MHz), not the ground stations (1 030 MHz). Everything on the page is inferred from the replies that reached the antenna in Tuusula. The figures in this guide show the page with *simulated* traffic — the layout is the real one, the numbers on your screen will differ.

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1 Mode S and ADS-B in a nutshell

1.1 Secondary surveillance radar and Mode S

Interrogation and reply. Secondary surveillance radar (SSR) does not see the aircraft by its echo: a ground station *interrogates* on 1 030 MHz and the aircraft's *transponder* replies on 1 090 MHz. The old modes A and C answer every interrogation with the identity code (squawk) or the pressure altitude; with many aircraft their replies overlap (*garbling*) and a radar also receives replies meant for other radars (*FRUIT*, false replies unsynchronised in time).

Mode S ("select") solves this by giving every aircraft a unique **24-bit ICAO address**. A ground station first finds new aircraft with an *all-call* interrogation, which every Mode S transponder answers with its address; then it asks each aircraft individually (*roll-call*), addressed to it alone [1, 3]. A station that has acquired an aircraft can *lock it out* of its own further all-calls, so that the aircraft does not keep answering them. Each station identifies itself in the all-call by an **interrogator code**: an *II code* (interrogator identifier, 0–15) or an *SI code* (surveillance identifier, 1–63, added in 1997 for denser networks) [5].

Squitters. Without being asked, a Mode S transponder also broadcasts its address about once a second (the *acquisition squitter*), and an ADS-B transponder broadcasts its position, velocity, identification and status several times a second (the *extended squitter*, Section 1.4). Anyone can receive these, including other aircraft (ACAS / TCAS) and passive receivers like the Radarcape.

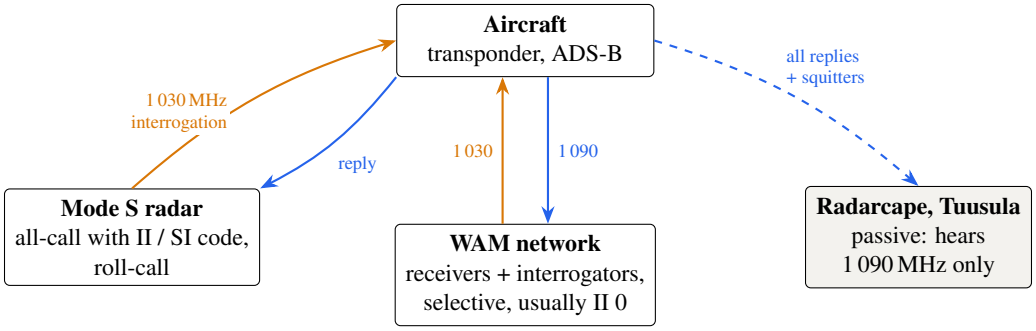


Figure 1. Who talks to whom. The ground stations interrogate on 1 030 MHz; the aircraft answers on 1 090 MHz and also broadcasts squitters. The Radarcape hears every reply and squitter that reaches Tuusula, whoever asked — but never the interrogations.

Around Helsinki the interrogations come mostly from Fintraffic's nationwide **wide-area multilateration (WAM)** network, together with Mode S radars [9, 10]. WAM locates the aircraft from the arrival times of its signals at several receivers; it also interrogates the aircraft selectively, as often as configured, and is not tied to the rotation of a radar antenna. Active WAM systems generally use II 0 [5]. The rate at which WAM asks an aircraft for its registers is the sample rate of the wind data of MODE-S Wind (Section 2.8).

1.2 The messages: downlink formats

A Mode S message is 56 bits (*short*, 64 µs with its preamble) or 112 bits (*long*, 120 µs), sent at 1 Mbit/s. The first five bits are the **downlink format** (DF); the last 24 bits are **parity** (Fig. 2).

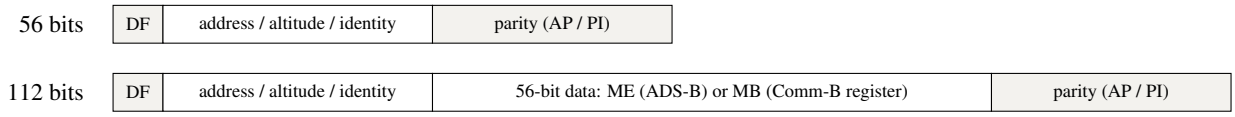


Figure 2. The two message lengths. In a long message the 56-bit data field carries one ADS-B message (DF17/18, field ME) or the content of one Comm-B register (DF20/21, field MB).

DF	Length	Content	Parity
0	short	Short air–air surveillance: reply to another aircraft's ACAS (TCAS)	AP
4	short	Surveillance reply, altitude (to a ground roll-call)	AP
5	short	Surveillance reply, identity (squawk)	AP
11	short	All-call reply, or the spontaneous acquisition squitter	PI
16	long	Long air–air surveillance (ACAS coordination)	AP
17	long	Extended squitter: ADS-B from a transponder	PI
18	long	Extended squitter from a non-transponder device; TIS-B, ADS-R	PI
19	long	Military extended squitter	—
20	long	Comm-B reply, altitude + 56-bit register content	AP
21	long	Comm-B reply, identity + 56-bit register content	AP
24	long	Comm-D extended length message	AP

Table 1. The downlink formats counted on the page.

Parity, and why some addresses cannot be verified. The parity is a 24-bit cyclic redundancy check (CRC) of the message. In **PI** formats (DF11/17/18) the CRC of a correct message is zero — or, in an all-call reply, the code of the interrogator that asked (Section 2.9). In **AP** formats (address/parity: DF0/4/5/16/20/21) the CRC is combined with the aircraft's address: the receiver recovers the address *from* the CRC, so a corrupted message simply yields a wrong address. The decoder (pyModeS [4]) therefore marks an AP reply as *verified* only when its address belongs to an aircraft already heard in its own squitters; the share of unverified Comm-B replies is shown on the page.

1.3 Comm-B and the registers

Ground-initiated Comm-B (GICB). A roll-call interrogation can ask the transponder for one of its 256 *registers* (Comm-B data selectors, BDS, written as two hexadecimal digits, e.g. 5,0). The transponder fills the registers from the flight computers and answers with a DF20 or DF21 reply carrying the 56 bits of that register (field MB). The reply does *not* say which register it is: the receiving ground station knows what it asked, a passive receiver does not. pyModeS therefore infers the register from the content — each register has fixed status bits, value ranges and consistency rules — and when the bits fit more than one register it picks the most likely one; the page shows the share of such *ambiguous* inferences.

ELS and EHS. Europe requires *Elementary Surveillance* (ELS: registers 1,0, 1,7, 2,0 and 3,0) for IFR flights, and *Enhanced Surveillance* (EHS: 4,0, 5,0, 6,0) together with ADS-B Out for aircraft heavier than 5 700 kg or faster than 250 kt [6]. Practically all airliners therefore answer the EHS registers, which carry the true airspeed, Mach number and heading that MODE-S Wind turns into wind and temperature [11].

BDS	Content (resolution)	Used by MODE-S Wind
1,0	Data link capability: transponder level, ACAS, Mode S subnetwork version	—
1,7	Common-usage GICB capability: which registers the aircraft supports	—
2,0	Aircraft identification: the callsign, 8 characters	callsign
3,0	ACAS active resolution advisory	—
4,0	Selected vertical intention: MCP / FCU selected altitude and FMS selected altitude (16 ft), barometric pressure setting (0.1 hPa)	—
4,4	Meteorological routine air report (MRAR): wind, static air temperature, pressure, turbulence, humidity — optional, rare	direct meteo
4,5	Meteorological hazard report (MHR): turbulence, wind shear, microburst, icing, wake vortex — optional, rare	hazards
5,0	Track and turn report: roll angle (0.18°), true track (0.18°), ground speed (2 kt), track angle rate, true airspeed (2 kt)	wind, windshear
6,0	Heading and speed report: magnetic heading (0.18°), indicated airspeed (1 kt), Mach (0.004), barometric and inertial vertical rates (32 ft/min)	wind, temperature

Table 2. The Comm-B registers counted on the page [2, 3]. The wind is the ground-velocity vector of 5,0 minus the air-velocity vector from the true airspeed and the heading of 6,0; the temperature follows from the Mach number and the true airspeed.

1.4 ADS-B: the extended squitter

An ADS-B transponder broadcasts DF17 messages by itself, without being interrogated. The first five bits of the 56-bit ME field are the **type code** (TC), which tells what the message carries (Table 3). An airborne aircraft sends its position and velocity about twice a second each, its identification about every 5 s, and its status messages every 1.25–2.5 s [3, 8].

TC	Message	Typical rate
1–4	Identification and category (callsign, wake category)	every 5 s
5–8	Surface position	on the ground
9–18	Airborne position with barometric altitude (the TC also encodes the position integrity)	2 per s
19	Airborne velocity: ground speed and track (subtypes 1–2) or airspeed and heading (3–4); vertical rate; the **GNSS − barometric altitude difference** (25 ft steps)	2 per s
20–22	Airborne position with GNSS height	rare
28	Aircraft status: emergency / priority, ACAS RA broadcast	on events
29	Target state and status: selected altitude, barometric pressure setting, autopilot modes	every 1.25 s
31	Operational status: ADS-B version, NIC / NACp supplements, capabilities	every 2.5 s

Table 3. ADS-B type codes (DF17).

Versions. The *ADS-B version* in the operational status message tells which standard the transponder follows: 0 (RTCA DO-260), 1 (DO-260A) or 2 (DO-260B / EUROCAE ED-102A, required by the European mandate) [7]. Version 2 transponders report the position accuracy and integrity in the form used by the GPS Quality pages (NACp, NIC).

The GNSS − baro difference in the velocity message is what MODE-S Wind uses for the heights of the pressure surfaces (Sounding and Wind Map guide). DF18 is the same message from a device that is not a transponder (e.g. an airport vehicle) or a rebroadcast by a ground station (TIS-B, ADS-R; mainly in the USA).

2 The Statistics page

Statistics is a link in the top bar. The page asks the server for the counts every 10 seconds (not while its browser tab is hidden). The Window buttons — 1 min, 10 min (default), 1 h and 24 h — choose the period of all tables, the signal histogram and the range rings; their rates are per second averaged over that period. The choice is remembered in the browser. To the right of the Window buttons are the **system health** indicators (Section 2.2). Most headers, column titles and values explain themselves when hovered.

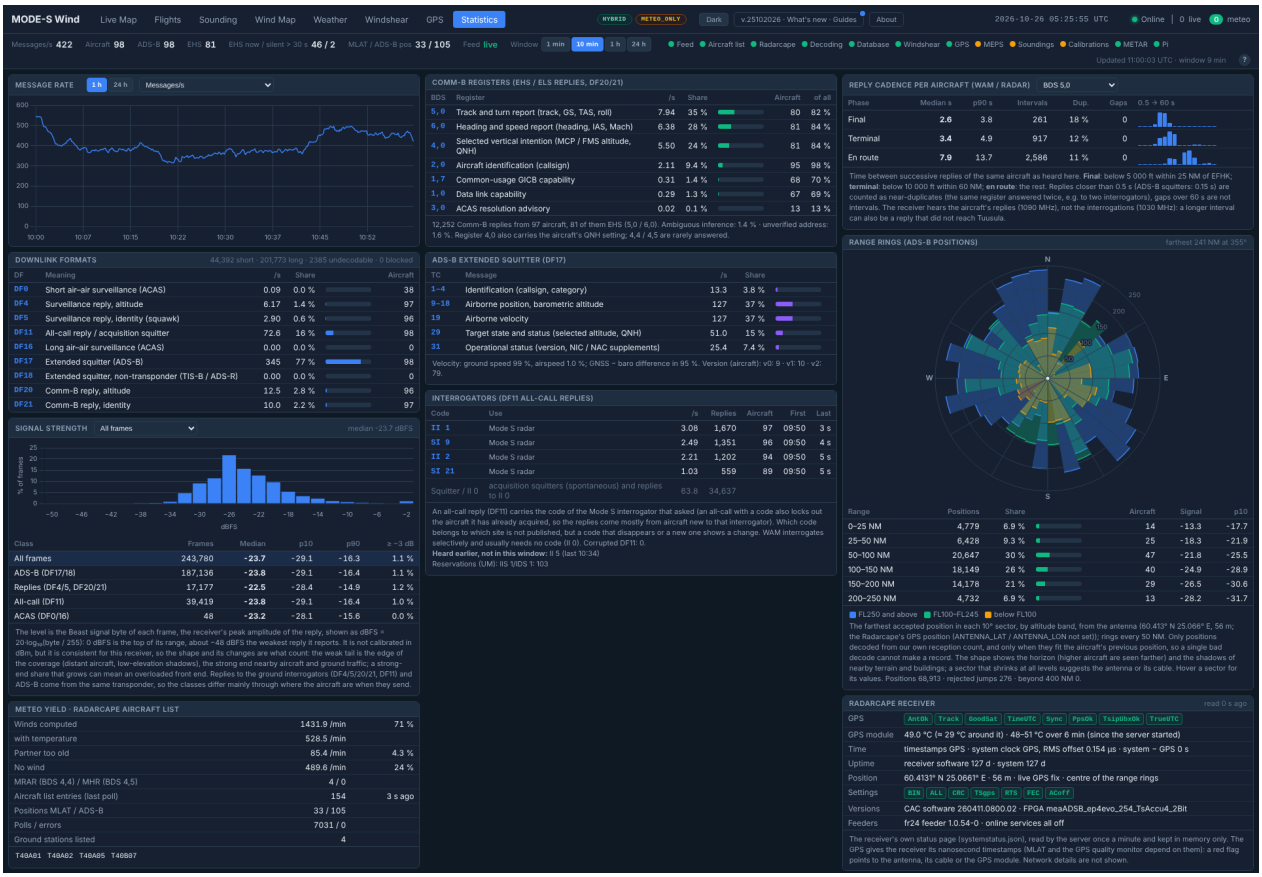


Figure 3. The Statistics page (simulated traffic). Left column: message rate, downlink formats, signal strength, meteo yield and the Radarcape aircraft list. Middle: Comm-B registers, ADS-B type codes, interrogators. Right: reply cadence, range rings, Radarcape receiver. On screens narrower than about 1 500 px the page has two columns, on small screens one.

How the counting works. The collector counts every frame from the Radarcape Beast feed before decoding, then each decoded message by format, type code and register, and the time since the same aircraft's previous message of the same kind. The counts are kept in memory — per-minute records for the last hour, the same merged per clock hour for 24 hours, and a compact per-minute series for the charts. Counting costs a few microseconds per message.

Kept over restarts. Each clock hour, once it has ended, is written as one compressed row to the database (about 10–50 kB; kept 7 days, `MSG_STATS_DAYS`). At a normal stop or restart of the server the hour in progress and the recent minute records and chart series are saved as well, and everything is read back at the start: the page continues where it was. The time the server was down is a *break* in the charts and is left out of the rates; the health panel lists the restart. After a crash or a power cut only the hour in progress is lost. All this is about 0.3–1 MB of writes a day.

2.1 Summary bar

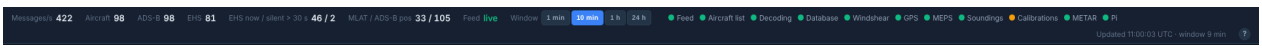


Figure 4. The summary bar.

- **Messages/s** — all frames of the last complete 10 seconds (hover: the average over the window). Typical daytime values are of the order of 1 000–2 000 per second.
- **Aircraft**, **ADS-B**, **EHS** — distinct addresses heard in the window; those that sent ADS-B (DF17); those that answered register 5,0 or 6,0.

- **EHS now / silent > 30 s** — aircraft heard in the last 10 s whose last BDS 5,0 reply is under 30 s old (being interrogated for EHS right now) / between 30 s and 10 min old (not interrogated for EHS at the moment, or their replies do not reach Tuusula). A rising "silent" count while traffic is normal points at a change in the ground system.
- **MLAT / ADS-B pos** — positions in the Radarcape aircraft list at the last poll, by source: multilateration of the Radarcape network, or the aircraft's own ADS-B.
- **Feed** — `live`, or `silent` with the time since the last frame (amber); the hover gives the connection uptime and the number of (re)connections since the server started.
- **Window** — the period of the tables; **Updated** at the right gives the time of the last refresh and the period actually covered (shorter just after the history begins).
- **Health dots** — one per part of the system (Section 2.2); on screens narrower than 1 600 px a single chip such as `Healthy 12/12`, `2 warnings` or `1 fault`.

2.2 System health

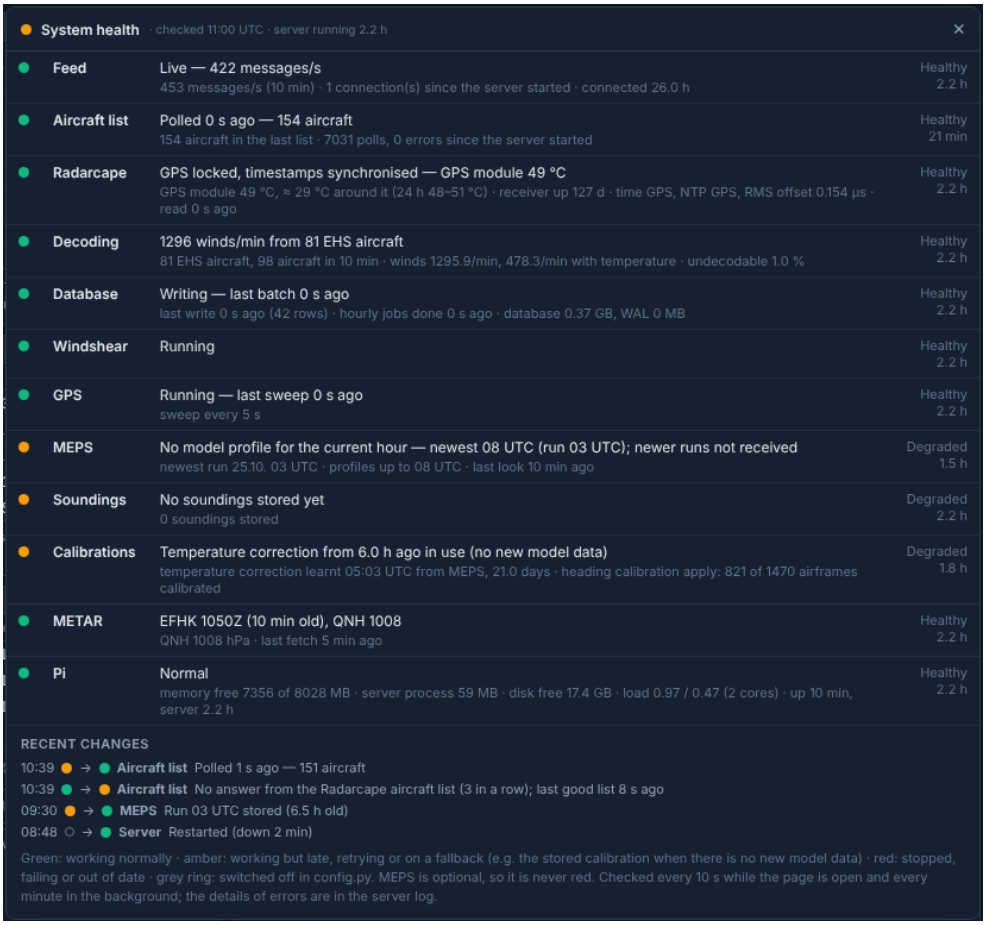


Figure 5. The system health panel (simulated): every part with its state, the reason in plain words, a few details and how long it has been in that state; below, the recent changes — here a restart, a model run that arrived, and a short break of the Radarcape aircraft list. The calibrations are amber because the temperature correction learnt from MEPS is 6 hours old: the system works on its stored values.

The dots right of the Window buttons show at a glance whether MODE-S Wind is working: **green** — working normally; **amber** — working, but late, retrying, or running on a fallback; **red** — stopped, failing or out of date; a **grey ring** — switched off in `config.py`. Hover a dot for its reason; click the row (or the chip) for the panel of Fig. 5, closed again with ×, Esc or a click outside.

Dot	Green when	Amber / red when
Feed	a message from the Radarcape within 5 s	silent 5–60 s / longer
Aircraft list	a good Radarcape aircraft list within 15 s	failing / for over 5 min
Radarcape	the receiver's GPS and timestamps in order, GPS module under 70 °C, settings as expected, status read within 5 min	a minor GPS condition, hot, clock off GPS time, software restarted, a setting differs / GPS fault, timestamps not locked, very hot, no answer for 15 min (Section 2.11)
Decoding	winds computed while EHS aircraft are heard (or none heard)	few winds or many undecodable messages / EHS aircraft but no winds for 10 min
Database	observations written; hourly jobs, purge check and memory clean-up running	database busy, a job late or failed / a write failed, or nothing written for 30 min
Windshear	the detection sweep running every 3 s	failing / not running for 2 min
GPS	the GPS quality sweep running	late / over 5 min
MEPS	newest model run at most 9 h old, and a model profile for the current hour	older, failing, or no profile within 1.5 h of now — newer runs not received (never red: the system works without the model)
Soundings	newest Jokioinen sounding at most 15 h old	15–30 h / older
Calibrations	temperature correction learnt from MEPS within 3 h	the stored correction in use (no new model data), or none learnt yet
METAR	METAR at most 75 min old (QNH for the low levels)	up to 3 h / older
Pi	CPU under 70 °C, power, memory, disk and load normal	hot, throttled, under-voltage seen, memory or disk low / CPU over 80 °C, under-voltage now, almost full

Table 4. The health indicators. A background task of the server that has stopped turns its dot red (… *stopped — restart the server*).

The checks run every 10 s while the page is open and every minute in the background, so the changes (the last 50, kept over restarts) are recorded also when nobody is watching; they are also written to the server log, where the details of any error are. An amber **Calibrations** or **MEPS** dot is not a fault to fix: it says that the system is working on its stored values because new model data has not arrived — by design MODE-S Wind does not depend on the model.

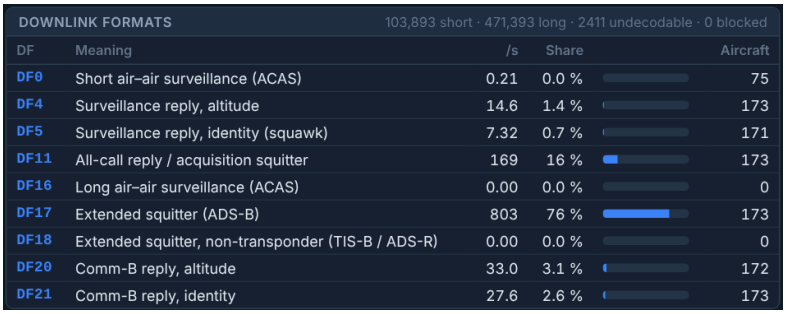
2.3 Message rate

The chart shows the messages per second over the last hour (one point per 10 s) or, with 24 h, per minute over the last day (kept over restarts; the time the server was down is a break in the line). The selector chooses the series of the chart: messages, the main formats (DF4/5, DF11, DF17, DF20/21), the registers 5,0, 6,0, 4,0 and 2,0, the numbers of aircraft and EHS aircraft per minute, the **median BDS 5,0 interval** on final, in the terminal area and en route (Section 2.8), and the number of interrogator codes heard.



Figure 6. 24-hour views (simulated): the messages per second follow the traffic of the day; the median BDS 5,0 interval is shortest on final and longest en route, and at night, with few landings, the value on final becomes noisy.

2.4 Downlink formats



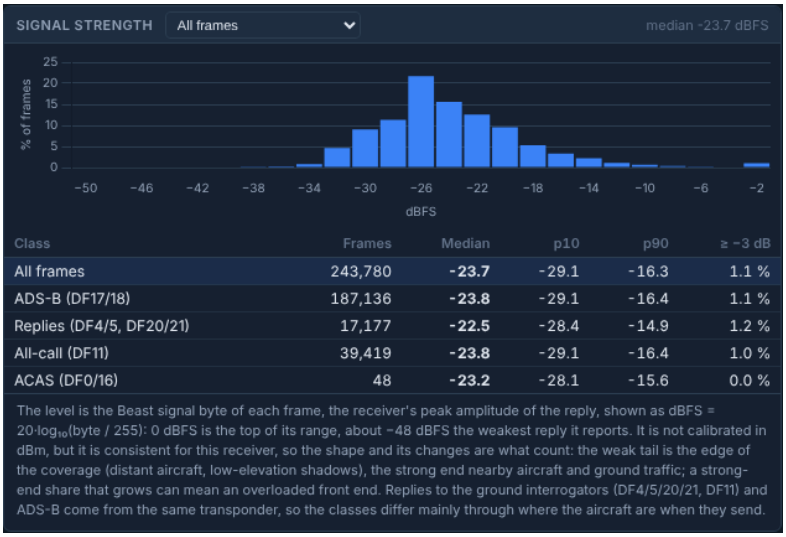
DOWNLINK FORMATS — 103,893 short · 471,393 long · 2411 undecodable · 0 blocked

DF	Meaning	/s	Share	Aircraft
DF0	Short air–air surveillance (ACAS)	0.21	0.0 %	75
DF4	Surveillance reply, altitude	14.6	1.4 %	173
DF5	Surveillance reply, identity (squawk)	7.32	0.7 %	171
DF11	All-call reply / acquisition squitter	169	16 %	173
DF16	Long air–air surveillance (ACAS)	0.00	0.0 %	0
DF17	Extended squitter (ADS-B)	803	76 %	173
DF18	Extended squitter, non-transponder (TIS-B / ADS-R)	0.00	0.0 %	0
DF20	Comm-B reply, altitude	33.0	3.1 %	172
DF21	Comm-B reply, identity	27.6	2.6 %	173

Figure 7. Downlink formats (simulated).

For each format: its meaning, the rate, the share of the decoded messages and the number of aircraft. The header gives the short and long frames, the undecodable ones (rejected by the decoder: corrupted, garbled or an unknown format) and the blocked ones (addresses on the ICAO blocklist). DF17 is usually the largest part (two position and two velocity squitters per second from every ADS-B aircraft), followed by DF11 (one acquisition squitter per second from every Mode S aircraft) and the replies to the ground interrogations (DF4/5 and DF20/21).

2.5 Signal strength



Class	Frames	Median	p10	p90	≥ −3 dB
All frames	243,780	**-23.7**	-29.1	-16.3	1.1 %
ADS-B (DF17/18)	187,136	**-23.8**	-29.1	-16.4	1.1 %
Replies (DF4/5, DF20/21)	17,177	**-22.5**	-28.4	-14.9	1.2 %
All-call (DF11)	39,419	**-23.8**	-29.1	-16.4	1.0 %
ACAS (DF0/16)	48	**-23.2**	-28.1	-15.6	0.0 %

The level is the Beast signal byte of each frame, the receiver's peak amplitude of the reply, shown as dBFS = $20 \cdot \log_{10}(\text{byte} / 255)$: 0 dBFS is the top of its range, about −48 dBFS the weakest reply it reports. It is not calibrated in dBm, but it is consistent for this receiver, so the shape and its changes are what count: the weak tail is the edge of the coverage (distant aircraft, low-elevation shadows), the strong end nearby aircraft and ground traffic; a strong-end share that grows can mean an overloaded front end. Replies to the ground interrogators (DF4/5/20/21, DF11) and ADS-B come from the same transponder, so the classes differ mainly through where the aircraft are when they send.

Figure 8. Signal strength (simulated): the share of frames in 2-dB steps, and per message class the median, the 10th and 90th percentiles and the share of strong frames.

The receiver reports for every frame its **signal level**, the peak amplitude of the reply as one byte (0–255). The page shows it as **dBFS** — decibels below the top of the receiver's range, $20 \log_{10}(\text{byte}/255)$, as dump1090 and readsb do: 0 dBFS is the strongest level the receiver reports, about −48 dBFS the weakest. The value is not calibrated in dBm (it depends on the antenna, the cable and the receiver's gain), but it is consistent for one receiver, so the *shape* of the histogram and its changes over days are what count.

The selector (or a click on a table row) shows one class: **all frames**, **ADS-B** (DF17/18), the **replies** to the ground interrogators (DF4/5, DF20/21), the **all-call** replies and squitters (DF11) or **ACAS** (DF0/16); *all frames* is then drawn as a line for comparison. The table gives per class the median, p10 and p90 and the share of frames within 3 dB of the top (≥ -3 **dB**).

- The **weak tail** is the edge of the coverage: distant aircraft and aircraft low on the horizon. If it moves to stronger levels while the traffic is the same, the receiver has become less sensitive (antenna, cable, connector, a new source of interference).
- The **strong end** is nearby aircraft and ground traffic. A share of strong frames that grows can mean an overloaded front end (a strong transmitter nearby, too much gain): strong replies are clipped and weak ones in between are lost.
- All classes come from the same transponders, so they differ mainly through *where* the aircraft are when they send: ADS-B is heard from all aircraft at all ranges, the replies of a busy terminal area come from closer aircraft.

The signal of the ADS-B *position* messages is also shown against the distance in the range-ring table (Section 2.10).

2.6 Comm-B registers

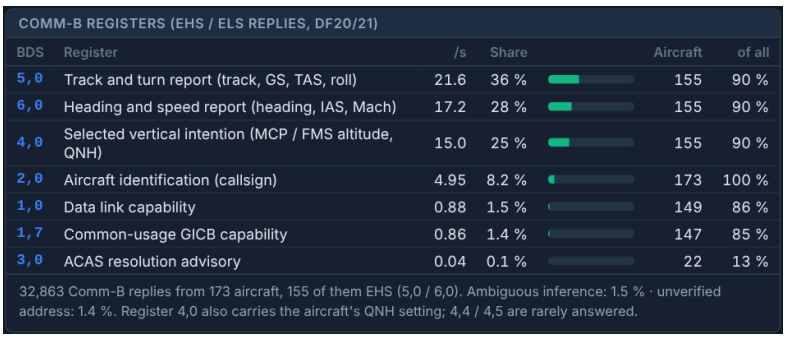


Figure 9. Comm-B registers (simulated).

Each register heard in the DF20/21 replies, with its rate, share of the Comm-B replies, the number of aircraft that answered it, and their share of all aircraft with Comm-B replies. The footer gives the share of ambiguous inferences and of replies with an unverified address (Section 1.2). The mix shows what the ground system is asking for: 5,0 and 6,0 dominate where WAM gathers EHS data; 2,0 (callsign) and 4,0 (selected altitude, used by controllers to see where an aircraft is going) come less often; 1,0 and 1,7 are asked rarely, as their content seldom changes.

2.7 ADS-B extended squitter

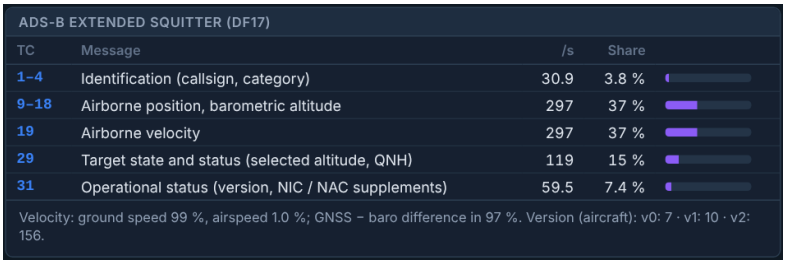


Figure 10. ADS-B type codes (simulated).

The type-code groups of Table 3 with their rates and shares. The footer gives the velocity subtypes (ground speed or airspeed), the share of velocity messages that carry the GNSS − baro difference, and the number of aircraft per ADS-B version (from TC 31).

2.8 Reply cadence per aircraft

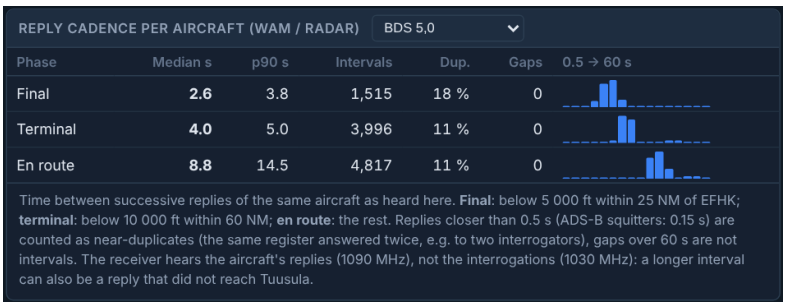


Figure 11. Reply cadence of BDS 5,0 (simulated): median, 90th percentile and histogram of the intervals per flight phase.

For the chosen register or message type — BDS 5,0, 6,0, 4,0, 2,0, the DF4/5 surveillance replies, the ADS-B position or velocity squitter — the table gives the time between successive messages of the *same aircraft*:

- per **flight phase**: **final** (below 5 000 ft within 25 NM of EFHK), **terminal** (below 10 000 ft within 60 NM), **en route** (the rest);

- **Median** and **p90** (90 % of the intervals are shorter), and a **histogram** from 0.5 to 60 s (hover a bar for its range and count);
- **Dup.** — *near-duplicates*: the same register again within 0.5 s (for the ADS-B squitters, which come every 0.4–0.6 s by themselves: within 0.15 s). They are not new samples; typically the aircraft answered two interrogators asking almost at the same time. They are left out of the intervals;
- **Gaps** — pauses of more than 60 s, not counted as intervals.

Why it matters. A wind needs a 5,0 and a 6,0 reply close in time, and the windshear detection follows TAS − GS from each 5,0 reply. The BDS 5,0 interval on final is therefore the time resolution of the approach winds and of the windshear detection: at a ground speed of 140 kt an interval of 2.5 s means one sample every 180 m. The ADS-B velocity squitter (every 0.5 s) gives the ground speed five times as often, but not the airspeed.

Reading it with care. The receiver only hears the replies. A long interval can mean that nobody asked, or that the reply did not reach Tuusula (terrain, distance, a weak or shadowed antenna on the aircraft, interference). Intervals of aircraft close to the receiver and high up are the most reliable measure of the interrogation schedule.

2.9 Interrogators

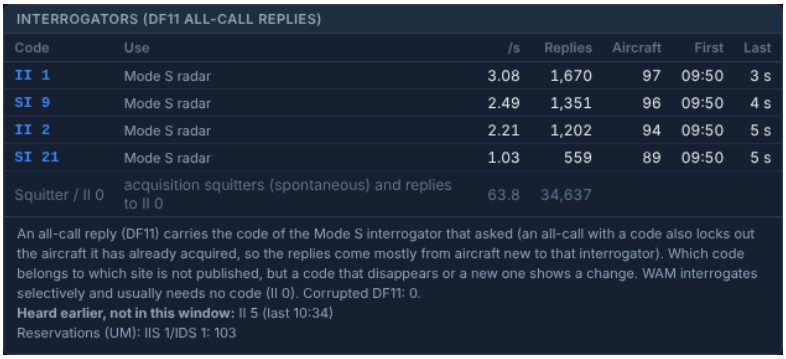


Figure 12. Interrogators (simulated codes; in the middle column under the ADS-B type codes): the codes heard in all-call replies with their use, rate and aircraft; below the table a code heard earlier but not in the last 10 minutes.

The parity of an all-call reply (DF11) is the CRC of the message combined with the code of the interrogator that asked. Computing the CRC therefore reveals **which interrogator** the aircraft answered — the only place where a passive receiver sees the ground side. A remainder of 0 is either the spontaneous acquisition squitter or a reply to II 0 (the two cannot be told apart); any other value that is not a valid code means a corrupted reply.

II	matching SI codes	Use (ICAO EUR / MID practice)
1–13	IC + 16, + 32, + 48	Operational Mode S radars; allocated by the EUROCONTROL MICA cell
0	16, 32, 48	Multilateration, WAM and mobile interrogators (not allocated)
14	14, 30, 46, 62	Test, research and development
15	15, 31, 47, 63	Military

Table 5. Interrogator codes [5]. An SI code has the same 4-bit value as its II code (e.g. II 1 and SI 1, 17, 33, 49).

For each code the table shows its use (Table 5), the reply rate, the replies in the window, the number of aircraft that replied, when the code was first heard (kept over restarts; with the date when it is older than a day) and how long ago its last reply came. Codes heard earlier but not in the window are listed below the table. Because a radar locks out the aircraft it has acquired, its all-call replies come mostly from aircraft new to it — entering its coverage, taking off, or recently changed — so a code's rate is not a measure of how busy the radar is, but its presence is a sign that it is transmitting. Which site uses which code is not published.

The footer also sums up the **reservation subfields** of the surveillance and Comm-B replies (UM: IIS, the code of the interrogator holding a multisite reservation, and IDS, its type). They are rarely non-zero.

2.10 Range rings

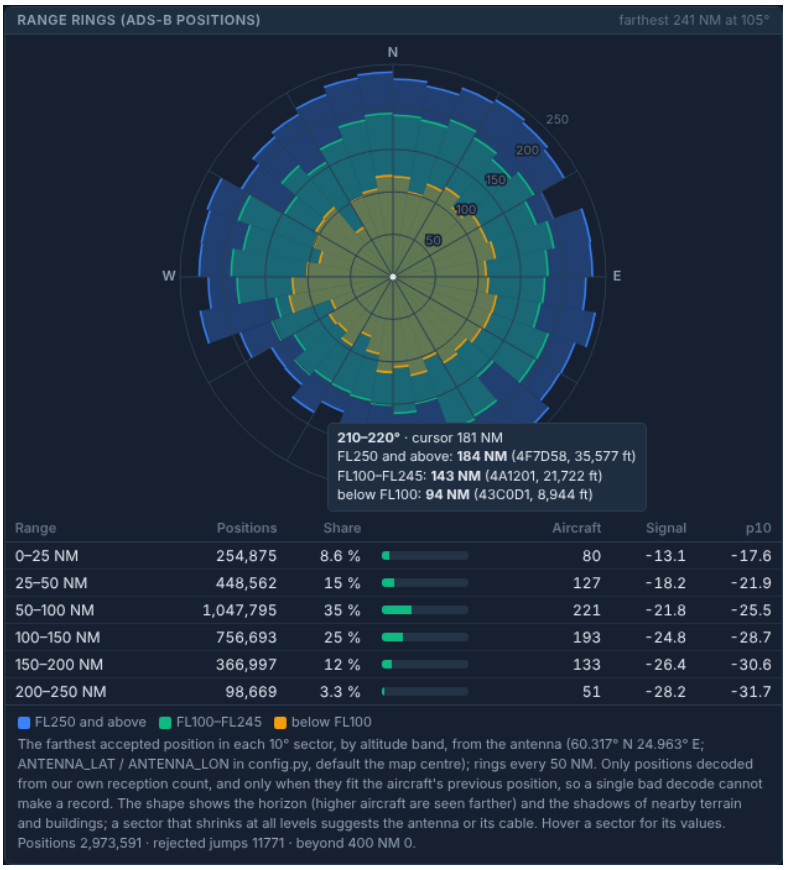


Figure 13. Range rings over 24 hours (simulated; in the right column under the reply cadence): the farthest position in each 10° sector for three altitude bands, with the hover of the 210–220° sector, and the positions, aircraft and signal per range band.

The range rings show the **coverage** of the receiver from the ADS-B positions that it decoded itself (airborne position messages, DF17 TC 9–18 and 20–22). For each **10° bearing sector** the plot shows the **farthest** position of the window, separately for three altitude bands: FL250 and above (blue), FL100–FL245 (green) and below FL100 (amber); the rings are every 50 NM (100 NM beyond 260 NM). Hover a sector for its three ranges with the aircraft and its altitude; the header gives the farthest position of the window.

A position counts only when it fits the same aircraft's previous one (at most 2 NM plus 0.35 NM per second apart, within a minute), so a single bad decode cannot set a record; positions beyond 400 NM are ignored. The centre is the antenna: the Radarcape's own GPS position while it is good — a live fix, antenna OK, three or more satellites, status read within 15 minutes (Section 2.11); otherwise `ANTENNA_LAT` / `ANTENNA_LON` in `config.py`, and without them the map centre (`RECEIVER_LAT` / `RECEIVER_LON`). The choice is made again at every status reading, so the centre falls back and returns by itself. The footnote names the source, the reason when the GPS position is not used, and the antenna height.

Reading the plot. A 1 090 MHz signal travels in a straight line, so an aircraft is heard only above the **radio horizon**, at most about

$$d \approx 1.23 \left(\sqrt{h_{\text{aircraft}}} + \sqrt{h_{\text{antenna}}} \right) \text{ NM} \qquad (h \text{ in feet}),$$

e.g. about 240 NM at FL380 and 120 NM at 10 000 ft for an antenna 30 ft up. The higher bands therefore reach farther, and the outline of each band is set by the horizon, by terrain, trees and buildings near the antenna (a sector that is short at all levels), and in some directions by the lack of traffic (e.g. over the sea or closed airspace). Below the plot the table gives per range band the positions, their share, the aircraft and the **median and p10 signal** of those position messages: the signal falls with the distance, and where it reaches the bottom of the receiver's range the coverage ends.

2.11 Radarcape receiver

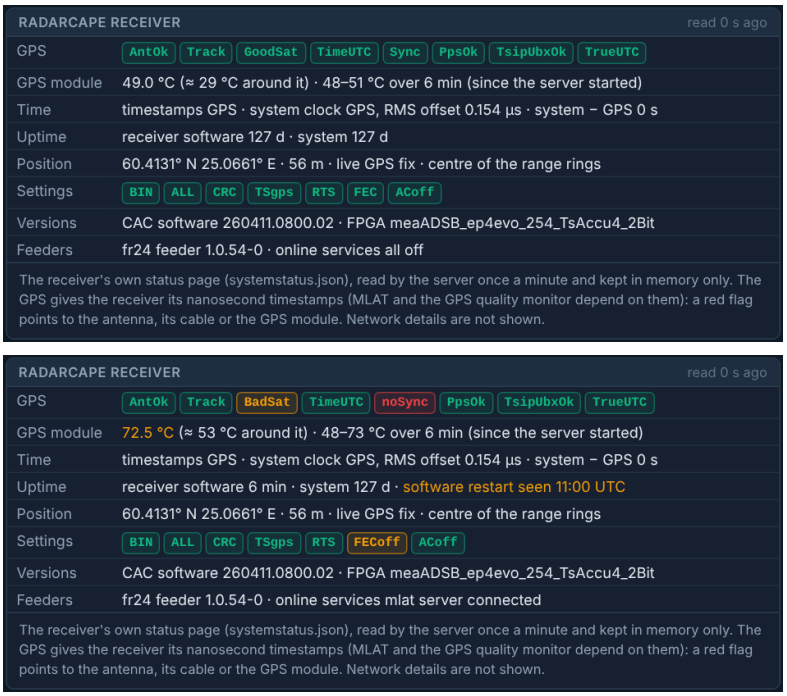


Figure 14. The Radarcape receiver panel (in the right column under the range rings). Top: all in order. Bottom (simulated faults): only one or two satellites (*BadSat*, amber), the timestamp clock not locked (*noSync*, red), a hot GPS module, a software restart a moment ago and the error correction switched off (*FECoff*).

The panel shows the receiver's **own status**: the Radarcape publishes it on its status page (`systemstatus.json`), which the server reads once a minute and keeps in memory only. Everything else on the Statistics page is about the messages; this panel is about the box that receives them.

GPS. A Radarcape is more than a radio: its GPS gives every message a timestamp accurate to nanoseconds, which multilateration (MLAT) needs and which the GPS quality monitor of MODE-S Wind relies on. The eight GPS flags are shown as chips (hover for the meaning):

Good	Otherwise	Meaning
AntOk	AntFail	GPS antenna connected and working (fail: short or open circuit — the antenna or its cable)
Track	notTrack	the GPS receiver tracks satellites
GoodSat	BadSat	three or more satellites (bad: only one or two)
TimeUTC	TimeGPS	the GPS–UTC offset is known, timestamps in UTC
Sync	noSync	the internal clock is locked to the GPS for nanosecond timestamps
PpsOk	PpsFail	the one-pulse-per-second signal of the GPS arrives
TsipUbxOk	TsipUbxFail	serial data from the GNSS module arrives
TrueUTC	Daysec+1	second-of-day counted from 0 at midnight (legacy: 1–86 400)

Table 6. The GPS flags of the Radarcape. Red: the timestamps cannot be trusted; amber: a minor condition, usually brief after a start.

After a power-up the flags turn green one by one within a few minutes. A red flag that stays points at the GPS antenna (sky view, snow or ice), its cable or the GPS module. Note that GNSS jamming in the area can also affect the receiver's own GPS — compare with the GPS Quality page.

GPS module. The temperature that the Radarcape reports is that of its GPS module, with the lowest and highest since the server started (up to 24 hours). The module runs about 20 K above the air around it [12], so the panel also gives that estimate: 49 °C on the module is about 29 °C inside the enclosure. It follows the weather and the enclosure but is not the air temperature. Amber from 70 °C, red from 85 °C (about 50 / 65 °C around the module).

Time. Where the message timestamps (*MLAT time source*) and the receiver's system clock take their time — normally both GPS — with the NTP RMS offset of the system clock as the receiver reports it (in microseconds), and the system clock minus the GPS time (normally 0 s).

Uptime. How long the receiver software and the receiver's operating system have been running. When the software uptime goes back while the server is running, the software has restarted: the panel and the health indicator show it for an hour.

Position. The receiver's GPS position and height, as a live GPS fix or a stored position. While it is good it is the centre of the range rings (Section 2.10); otherwise the panel says in amber *not used for the range rings*, why, and which position is in use. With `ANTENNA_LAT` / `ANTENNA_LON` set in `config.py` it also shows how far that position is from the receiver's own.

Settings. The receiver settings that MODE-S Wind relies on: Beast binary output (*BIN*), all downlink formats (*ALL*, not only DF11/17/18), CRC check (*CRC*), GPS timestamps (*TSgps*), the data-flow handshake (*RTS*, strongly required), the 1-bit error correction (*FEC*; off, about 15 % fewer frames) and Mode A/C off (*ACoff*). A setting that differs is amber. They are changed on the Radarcape's own web page.

Versions and feeders. The receiver software and FPGA versions (hover for the board and operating system), the feeder programs installed on the receiver and whether its online services are on. The network details of the receiver are not shown.

Health indicator. The **Radarcape** dot (Table 4) sums up the panel: red for an antenna fault, no satellites, timestamps not locked, 1PPS or GNSS data missing, a very hot module or no answer for 15 minutes; amber for the minor conditions. The reading can be switched off with `RADARCAPE_STATUS = False` in `config.py`; the dot is then grey.

2.12 Meteo yield and the Radarcape aircraft list



Figure 15. Meteo yield and the Radarcape aircraft list (simulated).

For every new BDS 5,0 or 6,0 reply: **winds computed** (the other register was close enough in time and the checks passed; with a Mach/TAS temperature as well), **partner too old** (the other reply was older than `WIND_MAX_AGE_SEC`: the wind would belong to another altitude and is not stored), **no wind** (no partner within `WIND_MAX_PAIR_AGE`, or rejected by the roll, track-rate or speed checks), and the rare MRAR / MHR reports. Below them: the entries of the Radarcape aircraft list at the last poll, its positions by source (MLAT / ADS-B), polls and errors, and the ground stations on the ICAO blocklist that the list contains (e.g. the `T40...` WAM stations, with their source and last time in the hover).

3 What you can analyse

3.1 The sample rate of the wind data

Choose BDS 5,0 in the cadence table and compare the phases. On final the median interval is the time resolution of the approach winds and of the windshear detection; en route it decides how many aircraft contribute to an hourly profile. With the `24 h` chart and the series *BDS 5,0 interval* the same values are seen through the day: a change at a fixed hour, or a step that stays, is a change of the interrogation schedule. Compare 5,0 with 6,0: if 6,0 is asked clearly less often, fewer 5,0 replies find a partner and the *no wind* share in the meteo table rises.

3.2 Changes in the ground system

- **Interrogator codes** that appear or disappear (the list below the table, or the series *interrogators heard* of the 24-h chart): a radar switched off for maintenance, a new one in service, a change of code allocation.
- **EHS silent > 30 s** rising while the traffic is normal: aircraft are not being asked for their EHS registers (or not heard), e.g. part of the WAM network out of service.
- **Near-duplicates**: their share shows how often two stations ask the same aircraft for the same register within half a second — overlapping coverage.

- **DF4/5 rate per aircraft**: the surveillance replies show how often the aircraft are interrogated at all; with the EHS cadence it separates "not interrogated" from "interrogated, but not for EHS".

3.3 The aircraft fleet

The share of EHS aircraft among the Comm-B aircraft, the ADS-B versions and the share of velocity messages with the GNSS − baro difference describe the fleet over Southern Finland: business jets and smaller aircraft answer the ELS registers but often not EHS, older transponders report ADS-B version 0 or 1. MRAR (4,4) and MHR (4,5) replies show whether any aircraft in the area deliver direct meteorological reports — at Helsinki-Vantaa practically none do, which is why MODE-S Wind computes wind and temperature from 5,0 and 6,0.

3.4 Receiver and data quality

- **Messages per second** through the day follow the traffic; a sudden drop with normal traffic points at the receiver, the antenna or the network.
- **Undecodable share** (header of the downlink formats): high values mean interference or heavy garbling.
- **Feed** state and reconnections: interruptions of the Beast connection.
- **MLAT vs ADS-B positions**: when GNSS interference removes the ADS-B positions of many aircraft, the MLAT share rises — compare with the GPS Quality and GPS Episodes pages.
- **Unverified Comm-B addresses**: mostly replies from aircraft not yet heard in their own squitters, or corrupted replies.
- **Signal strength**: compare the histogram of today with the same time of another day. A weak tail that has moved to stronger levels, or a strong-end share that has grown, points at the antenna, the cable or an overloaded receiver (Section 2.5).
- **Range rings**: over 24 hours the outline is a stable picture of the coverage. A sector that shrinks at all levels points at a new obstacle or the antenna; all sectors shrinking together, at the receiver or the cable. Use the `24 h` window, as one hour may simply have no traffic in some directions.
- **Radarcape receiver**: the GPS flags all green and the timestamps synchronised; the GPS module temperature over the day and the seasons; an unexpected software restart or a changed setting (Section 2.11).
- **Health dots**: a quick check that the feed, the decoding, the database and the background jobs are running, and how the Raspberry Pi itself is doing (temperature, power, memory, disk).

3.5 ACAS activity

DF0 and DF16 are replies to the collision avoidance systems of other aircraft, register 3,0 and ADS-B TC 28 carry ACAS resolution advisories. Their rates show the ACAS interrogation load in the area; a 3,0 reply with content, or a TC 28 RA broadcast, marks an actual resolution advisory.

Question	Where to look	What to expect
How often are the winds sampled on final?	Cadence, BDS 5,0, Final	median about 2–3 s
Did the interrogation schedule change?	24-h chart, *BDS 5,0 interval*	a step or an hourly pattern
Is a radar out of service?	Interrogators, *heard earlier*	a code missing for hours
Why fewer winds than usual?	Meteo yield, cadence of 6,0	*no wind* share up
Is the receiver all right?	Messages/s, undecodable, Feed	steady with traffic
Is the antenna all right?	Range rings and signal, 24 h	outline and weak tail as on other days
Is the receiver's GPS all right?	Radarcape receiver, GPS	all flags green, *Sync*
Is the system working?	Health dots	all green (amber Calibrations / MEPS: on stored values)
GNSS interference now?	MLAT / ADS-B positions	MLAT share up

Table 7. Quick analyses.

4 Limitations

- **One receiver.** Everything is seen from Tuusula: coverage, terrain and the aircraft antenna decide what is heard. Low aircraft far away are not heard; their intervals look longer than they are.
- **Replies, not interrogations.** The schedule of the ground stations is inferred from the replies; an unanswered or lost interrogation is invisible.
- **Inferred registers.** The register of a Comm-B reply is inferred by the decoder; a small share is ambiguous or wrong.
- **Codes, not sites.** Interrogator codes are known, their sites are not published; II 0 replies cannot be separated from acquisition squitters.

- **History.** Kept over restarts (7 days in the database, 24 hours on the page); after a crash or a power cut the hour in progress is lost.
- **Signal not in dBm.** The signal level is relative to the receiver's range; compare it only with itself.
- **Receiver status as reported.** The Radarcape panel shows what the receiver says about itself, once a minute; the temperature is that of the GPS module (about 20 K above the air around it), not of the whole box or the outside air.
- **Range from ADS-B only.** The range rings use the ADS-B positions decoded by this receiver; aircraft without ADS-B (or with GNSS interference) do not contribute.

5 Quick reference

Item	Meaning
Window	period of the tables: 1 min, 10 min, 1 h, 24 h
Messages/s	frames of the last 10 s
EHS	aircraft that answered 5,0 or 6,0 in the window
EHS silent > 30 s	heard now, last 5,0 reply 30 s – 10 min ago
Ambiguous	Comm-B replies whose bits fit more than one register
Unverified	AP replies whose address is not known from squitters
Median / p90	interval between successive replies of the same aircraft
Dup.	same register again within 0.5 s (ADS-B: 0.15 s; not a new sample)
Gaps	pauses over 60 s
II / SI	interrogator codes from DF11 all-call replies
Squitter / II 0	spontaneous acquisition squitters and replies to II 0
Winds computed	new 5,0 / 6,0 replies that completed a wind
Partner too old	wind not stored: the other reply too old
dBFS	signal level below the top of the receiver range, $20\log_{10}(\text{byte}/255)$
≥ -3 dB	strong frames, near the top of the receiver range
Range rings	farthest accepted ADS-B position per 10° sector and altitude band
Health dots	green working, amber late / retrying / on a fallback, red stopped or failing, grey off
GPS flags	the Radarcape's GPS: antenna, tracking, satellites, UTC, *Sync* (timestamp clock locked), 1PPS, module data
History	hourly rows kept 7 days (`MSG_STATS_DAYS`); restarts are breaks in the charts
API	`/api/msgstats?window=1...60` or `1440`; `/api/health`

References

[1] ICAO. *Annex 10 to the Convention on International Civil Aviation, Aeronautical Telecommunications, Volume IV: Surveillance and Collision Avoidance Systems*.

[2] ICAO. *Technical Provisions for Mode S Services and Extended Squitter* (Doc 9871), 2nd edition, with amendments.

[3] Sun, J. (2021). *The 1090 Megahertz Riddle: A Guide to Decoding Mode S and ADS-B Signals*. TU Delft OPEN Books. doi:10.34641/mg.11

[4] Sun, J., Vû, H., Ellerbroek, J., Hoekstra, J. M. (2020). pyModeS: Decoding Mode-S surveillance data for open air transportation research. *IEEE Trans. Intell. Transp. Syst.* 21(7), 2777–2786. doi:10.1109/TITS.2019.2914770

[5] EUROCONTROL MICA / ICAO MID (2021). *Operation on II and SI codes*. MICA-MID 2021 webinar, WP03.

[6] Commission Implementing Regulation (EU) No 1207/2011 on the performance and interoperability of surveillance for the single European sky, as amended by (EU) No 1028/2014, 2017/386 and 2020/587; replaced in 2023 by Regulation (EU) 2023/1770.

[7] RTCA DO-260B / EUROCAE ED-102A. *Minimum Operational Performance Standards for 1090 MHz Extended Squitter ADS-B and TIS-B*.

[8] Bayliss, E. T., Boisvert, R. E., Burrows, M. L., Harman, W. H. (1994). Aircraft surveillance based on GPS position broadcasts from Mode S beacon transponders. *Proc. ION GPS-94*. MIT Lincoln Laboratory.

[9] ATC Network (2012). Finavia enhances surveillance capabilities through COMSOFT WAM network. 13 March 2012.

[10] Frequentis (2022). Finland transforms air traffic surveillance with Frequentis' highly resilient Wide Area Multilateration system. Press release, 20 June 2022.

[11] de Haan, S. (2011). High-resolution wind and temperature observations from aircraft tracked by Mode-S air traffic control radar. *J. Geophys. Res.* 116, D10111. doi:10.1029/2010JD015264

[12] Jetvision. *Radarcape: Software Features* (systemstatus.json fields). wiki.jetvision.de, accessed 26 October 2026.